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The role of intellectual property on Enterprise-led Industry-University-Research Institution cooperation: case of innovation awards in China

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The Enterprise-led Industry-University-Research Institution (E-IUR) cooperation innovation system is an integral component of the National Innovation System (NIS). This article explores the impact of intellectual property (IP) protection policies on E-IUR cooperation. Utilizing the achievement of prestigious innovation awards, this article analyzes the degree of E-IUR cooperation reflected in the ranking of the winning organizations, showcasing the position of the enterprises program in high-level technological innovation in China and the leadership among E-IUR cooperation. Additionally, this paper further explores the impact of intellectual property demonstration cities (IPDC) policy on E-IUR cooperation using a difference-in-differences (DID) regression model. The study focuses on listed companies in Shanghai and Shenzhen from 2008 to 2020. The results show that IPDC policy offers significant assistance in boosting the deep integration of E-IUR cooperation. Furthermore, refining the fundamental institution of IP protection has the potential to propel the prevalence of E-IUR cooperation. The IPDC policy has strengthened the deep integration of E-IUR cooperation by increasing enterprise innovation investment and regional IP law enforcement. This study investigates the dominant-subordinate relationship among multiple innovation organizations in IUR cooperation, measures the dominant degree of enterprises in IUR cooperation compared with universities and research institutions. And we empirically test the influence of IP on the deep integration of E-IUR cooperation, which provides a direction for optimizing the IP basic system and improving the overall efficiency of NIS and RIS in emerging economies.

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Introduction

The National Innovation System (NIS), first proposed by Freeman in 1987, is an institutional network composed of both the public and private sectors, to facilitate the emergence and development of new technologies (Freeman, 1987). NIS is a complex system consisting of a diverse range of participants, including enterprises, universities, and research institutions, with the ultimate objective of fostering the enhancement of the country's innovation capacity (Lundvall, 1992; Nelson, 1993; Erzurumlu et al., 2022; Weerasinghe et al., 2024). As primary innovators in a market economy, enterprises occupy a pivotal position and exert a significant influence on the NIS. In the context of China, the NIS has undergone a shift from government-led innovation to enterprise-led innovation (Sun and Liu, 2010; Băzăvan, 2019; Xu and Chen, 2023). The *Outline of the National Medium-and Long-Term Development Plan for Science and Technology (2006–2020)* points out that the NIS is a social system that is led by the government, fully utilizes the basic role of the market in resource allocation, and has close ties and effective interactions among various scientific and technological innovation programs. The Report to the 20th National Congress of the Communist Party of China (CPC) emphasized that enterprises are the mainstay of scientific and technological innovation, and strengthening the deep integration of industry-university-research innovation cooperation led by enterprises (E-IUR cooperation) is crucial to activate scientific and technological innovation resources and improve the efficiency of the NIS. In addition to the NIS, the Regional Innovation System (RIS) approach highlights the role of geographic regions in innovation, emphasizing localized interactions between enterprises, universities, and research institutions (Tödtling and Trippel, 2011, 2013). And the Sectoral Innovation System (SIS) framework focuses on sector-specific innovation, highlighting the interactions between enterprises, research institutions, and policymakers that drive technological advancements and production (Malerba, 2002). Together, the NIS, RIS, and SIS frameworks acknowledges the non-linear and multidisciplinary characteristics inherent in the innovation process, emphasizing the interactions and interplays among diverse actors who demonstrate a vested interest in innovation (Malerba, 2002; Watkins et al., 2015).

The significant role of IP strategy in the development of a country's innovation system has been widely acknowledged (Song, 2013; Watkins et al., 2015; Neves et al., 2021). Among numerous systems that aim to stimulate innovation and development, IP strategy plays an active role in driving technological innovation, promoting industrial innovation, and improving the innovation environment, and it also plays a decisive role in improving the competitiveness of innovative actors. The basic system of IP rights and the IUR cooperation are integrated into a unified framework, and it is of great theoretical foundation and practical significance to study the impact of IP protection policies on strengthening the deep integration of E-IUR cooperation (Siltaloppi and Ballardini, 2023).

Many researchers have investigated the relationship between IP protection and enterprise innovation (Brüggemann et al., 2016; Song et al., 2023; Zhou and Guo, 2023; Zheng et al., 2023; Song et al., 2024). Currently, there is no unanimous agreement on whether IP protection hinders or encourages innovation. However, numerous studies have adopted a favorable position, arguing that robust IP protection positively impacts enterprise innovation (Neves et al., 2021). It is important to note that the influence of IP protection on firm-level innovation outcomes is contingent on various factors, including the type of firms and countries (Papageorgiadis and Sharma, 2016; Neves et al., 2021). Bondarev (2018) asserts that the impact of IP protection varies for multinational enterprises and local businesses, indicating that

enhanced protection may foster innovation in large multinational corporations but impede it in the case of local businesses. Thakur-Wernz, Wernz (2022) found that the enterprises founded prior to the implementation of trade-related aspects of IP protection tend to exhibit lower levels of innovation awareness and focus on process innovation, while the firms established post-implementation demonstrate heightened innovation awareness, with a notable emphasis on product innovation. Song et al. (2023) evaluated the impact of IP policy on enterprise innovation and found that the implementation of IP policy significantly stimulated enterprise innovation investment and innovation quality. Zheng et al. (2023) found that increased judicial safeguards for IP positively influence digital innovation in enterprises. In particular, smaller firms demonstrate a higher susceptibility to the effects of judicial protection of IP on their digital innovation initiatives compared to larger enterprises.

Some researchers focus on the influence of IP protection on IUR cooperation performance. Arora and Gambardella (2010) believe that complete IP protection can promote R&D cooperation and technology transfer among organizations, which is conducive to the dissemination of knowledge. Okamuro and Nishimura (2013) posit that equitable and flexible university IP policies, introduced in Japan in 2003, enhance IUR collaboration project performance by increasing firm commitment, and this hypothesis is partially supported by empirical data from a survey of Japanese firms. George and Tarr (2023) studied the policy-making behind the Australian online standard contract, known as the IP Toolkit, through the lens of relational contract theory. They evaluated the impact of the new IP framework on fostering IUR cooperation. Nguyen et al. (2023) found that the demand of enterprises for IP rights is positively related to the tendency and degree of enterprises to carry out open innovation. Shi et al. (2020) suggested that the enforcement of IP rights can regulate the cooperation and innovation efficiency of enterprises in IUR cooperation, but the precise direction of this influence remains uncertain. IP rules can play a role as an adhesive to avoid harming innovative efficiency in IUR cooperation. However, although China's basic IP system provides basic principles for the acquisition and sharing of IP rights, it is not aimed at IUR innovation cooperation and the rules for application of the system are unclear.

The above research investigates the influence of the IP system on enterprise innovation or IUR innovation cooperation, but ignores that enterprises are in the NIS and innovation is the result of complex interaction among enterprises, universities, and research institutions in the NIS. IUR innovation cooperation research usually assumes that the relationship between multiple innovative organizations is equal or symmetrical, but, in fact, the power allocation caused by resource dependence is not balanced (Davis and Cobb, 2010; Hoppmann, 2021), and there is a typical dominant-subordinate relationship between cooperative organizations. This paper primarily addresses the impact of IP protection policies on E-IUR cooperation using a stagger difference-in-differences (DID) regression model. The results show that the establishment of IPDC offers significant assistance in strengthening the deep integration of E-IUR cooperation. Furthermore, refining the fundamental institution of IP protection has the potential to propel the prevalence of E-IUR cooperation.

This study makes a significant contribution to the literature on IP protection and E-IUR cooperation by offering three distinct and contextually grounded insights. First, most studies measure IUR cooperation by R&D investment, patent, or paper cooperation (Brüggemann et al., 2016; Fang et al., 2017; Song et al., 2023), ignoring the commercialization process of innovation. This study uses innovation awards as a proxy for the effectiveness of IUR

cooperation. Winning a prominent innovation award provides credible validation about innovation and the ability to observe the contribution of each organization in the cooperation process. This study focuses on award-winning innovations certified by third parties, which assess technological innovation, significant economic benefits, and industry advancement, providing a benchmark for expectations in E-IUR cooperation.

Second, while the existing literature has predominantly focused on the impact of IP protection on enterprise innovation (Brüggenmann et al., 2016; Song et al., 2023; Zhou and Guo, 2023; Zheng et al., 2023; Song et al., 2024) or IUR's innovation cooperation (Okumuro and Nishimura, 2013; Shi et al., 2020; George and Tarr, 2023), it has often overlooked the intricate interplay that drives innovation as a collective outcome. Existing collaborative research within the IUR framework often presumes an egalitarian or symmetric relationship among various innovation organizations, this study challenges this assumption. It acknowledges and addresses the overlooked issue of power imbalances arising from resource dependence dynamics (Song, 2013; Hoppmann, 2021). From the perspective of asymmetric dependence, this research offers a nuanced understanding of power dynamics within IUR cooperation by examining dominant-subordinate relationships among these organizations involved in innovation.

Finally, this study has integrated IP protection and the IUR innovation cooperation into the same framework and explores the influence of IP protection on the relationship among multiple innovation organizations in the RIS (Tödtling et al., 2022) and SIS (Malerba, 2002), enriching the research on the innovation effect of IP protection (Bondarev, 2018; Song et al., 2023; Nguyen et al., 2023). Existing research focuses on the impact of IP rights on R&D investment and patent applications (Brüggenmann et al., 2016; Fang et al., 2017; Song et al., 2023). In fact, IP protection holds greater practical value for innovations with substantial commercialization potential. The findings offer strategic guidance on how to enhance deep integration within the E-IUR cooperation model, specifically from the point of view of the institutional environment. This research not only provides empirical evidence of the benefits derived from IPDC policy but also offers actionable insights for policymakers and business leaders to refine the institutional infrastructure surrounding IP.

The remainder of this paper is organized as follows. In “2 Theoretical Background and Research Hypotheses,” we show theoretical background and institutional environment and research hypotheses. In “3 Research Design” sample selection and data sources, empirical strategy, and variables are presented. In “4 Results and Analysis” descriptive statistics, results for hypothesis testing, robustness test, and mechanism analysis are provided. In “5 Conclusions and Discussion” conclusions and policy implications are presented.

Theoretical background and research hypotheses

Theoretical background and institutional environment. The Global Innovation Index Report 2022, released by the World IP Organization and Cornell University, indicates that China's innovation index ranking has advanced significantly, moving from 43rd place in 2010 to 11th place in 2022, making it the sole middle-income economy to break into the top 30 (Li and Xiao, 2023). The International Property Rights Index 2023 shows that China's IP rights score ranking has increased from 63rd in 2010 to 28th in 2022. Although the development of IP rights is not as good as innovation, the Chinese government has made great efforts in this regard, such as the formulation of the “*Intellectual Property Powerful Country Construction Framework (2021–2035)*” and the IPDC policy construction, etc.

The IPDC policy is an important part of China's national strategic implementation of IP, which aims to improve the ability of demonstration cities to create, gives full role to the IP protection row in sustainable development driven by innovation (Zhou and Guo, 2023; Jiang et al., 2023). Specifically, the IPDC policy develops the culture of IP and further raises the awareness of IP protection of enterprises, universities, and research institutions. Meanwhile, the IPDC policy plays a pivotal role in strengthening IP protection by enhancing enforcement measures, bolstering management practices, and facilitating IP financing. Consequently, spurious and invalid patents are eliminated, contributing to an elevation in the overall quality of innovation. Theoretically, the implementation of IPDC policy has the potential to create a mutually beneficial scenario, fostering the concurrent development of innovation and collaboration in the realm of E-IUR cooperation.

Since the central government released the *Outline of the National Intellectual Property Strategy* in 2008, IP protection has been strengthened nationwide. To promote the establishment and implementation of IPDC policy in enterprises, universities, and research institutions, China National IP Administration (CNIPA) formulated the *Measures for the Assessment of National Pilot and Demonstration Cities (Urban areas) of Intellectual Property* in 2011. The primary goal of the IPDC is to enhance overall IP capabilities, develop and execute urban IP strategies, enhance management and service capabilities related to urban IP, enhance urban IP creation, and improve the efficacy of IP law enforcement to boost economic gains from IP transformation. In 2012, CNIPA recognized 23 cities, including Wuhan and Guangzhou, as demonstration cities for the first time in China. The selection process involves local governments submitting free applications, with CNIPA determining the final list of pilot cities. The IPDC initiative represents a novel project in IP, showcasing the voluntary commitment of local governments to safeguard property rights. Following initial recognition in 2012, CNIPA subsequently announced additional batches of demonstration cities in 2013, 2015, 2018, and 2019, further emphasizing the ongoing commitment to IP protection at the regional level. By December 2022, the CNIPA had approved a total of 77 national IP demonstration cities (urban areas), including 15 sub-provincial cities, 53 prefecture-level cities (districts), and 9 county-level cities.

Research hypotheses

The direct influence of IPDC policy on E-IUR cooperation. The close interaction of enterprises and universities, and research institutions is pivotal in sustaining NIS and RIS (Tödtling and Tripl, 2011, 2013) and is essential for ‘catching up’ emerging economies operating away from the global technological frontier (Vlasova, 2021). IP is among the most complex issues in joint R&D efforts among enterprises, universities, and research institutions. IP protection has consistently been seen as a crucial tool to encourage IUR cooperation (Panagopoulos, 2003). A robust IP protection system serves as a crucial prerequisite for the participation of enterprises and other innovative organizations, ensuring that participation in innovation competition is characterized by fairness and the assurance of reasonable returns from inventive endeavors (Yang et al., 2023).

First of all, in the process of IUR cooperation, different organizations (enterprises, universities, and research institutions) follow different institutional logics (Greenwood et al., 2011), especially in commercial logic. Universities and research institutions often prioritize the disclosure and dissemination of scientific knowledge as a public product, while enterprises may prefer knowledge protection such as patents, to avoid the problem of

“hitchhiking” (Nasirov and Joshi, 2023). This difference in commercial logic is likely to lead to IP conflicts among multiple innovation organizations. Undeniably, insufficient protection of IP undermines the motivation for enterprises to invest in R&D (Zhang and Shan, 2023; Jiang et al., 2023). In the absence of IP protection, the replication of innovation can be legally commercialized, transforming the innovation into a societal benefit (Li and Xiao, 2023). It is evident that without the safeguard of IP, competitors can imitate freely, leading to diminished profits for the originator. Consequently, the innovator may refrain from investing resources in innovative activities, thereby hindering prioritization in technological innovation. However, if local governments actively carry out the construction of IPDC and introduce relevant policies to effectively protect the IP rights of innovative programs, it will help innovative organizations to be more willing to disclose their R&D information, which alleviates the information asymmetry among all partners, effectively relieve the dilemma related to innovative information (Fang et al., 2017), and achieve the optimal overall efficiency of IUR cooperation.

Secondly, IPDC policy can promote the deep integration of IUR cooperation through the construction of the IP public service platform. As an important part of the Chinese government’s IP strategy, IPDC policy aims to improve the level of IP protection in demonstration cities. In the process of implementation, clear and strict assessment requirements and standards have been established, and the evaluation indicators for the construction of demonstration cities have been clarified in the Index System for Evaluating Demonstration Cities issued together with the *Measures for the Assessment of National Pilot and Demonstration Cities (Urban areas) of IP*. For example, at the macro level, the overall deployment of IP protection has been carried out, as well as the construction and operation of the overall coordination mechanism for IP. By combining the relevant policies and work plans of IPDC policy, we can find that building a public service platform for IP rights is an important part of the construction of IPDC. The construction of IPDC facilitates the profound integration of IUR cooperation through the proactive establishment of innovative R&D and technology business incubator platforms, formulation of IUR cooperation platforms encompassing governments, universities, enterprises, and research institutions, as well as the establishment of diverse technological innovation service platforms. Public service platforms can the capacity of the main body through government subsidies and mobilization of human resources, optimize the environment by improving local IP protection and reducing internal and external transaction costs, and establish a network by promoting IUR cooperation.

Finally, IPDC policy can improve the leading position of enterprises in IUR cooperation. A perfect basic system of IP rights can provide a better legal framework and protection and institutional support for enterprises, help enterprises to better safeguard their IP rights and interests, and improve their voice and influence in IUR cooperation. Enterprises have invested a lot of resources in R&D and IUR innovation cooperation, and the establishment of IPDCIP can ensure the legitimacy and sustainability of these efforts to a greater extent. Through the protection of IP rights in the form of patents, trademarks, copyrights, etc., enterprises can ensure that their unique innovation achievements will not be copied by others at will, thus having more motivation to invest in R&D. The imperfect protection mechanism of IP rights and achievements is one of the reasons why the dominant position of collaborative innovation decision-making of enterprises is not fully exerted. Enhancing IP protection across different regions fosters the development of pioneering enterprises in innovation (Jiang et al., 2023). Through the protection of IP rights, the innovative achievements produced

by enterprises in cooperation can bring economic returns and market competitive advantages to enterprises. This protection encourages enterprises to cooperate more actively with universities and research institutions to jointly promote scientific and technological innovation and improve industrial competitiveness. The cultivation of such innovative enterprises serves to fortify their dominant position, reinforcing their pivotal role in driving innovation within the industry. This, in turn, facilitates the implementation of an innovation-driven development strategy. On the basis of the above analysis, we propose the first hypothesis as follows.

Hypothesis 1 (H1): The IPDC policy is helpful to strengthen the deep integration of E-IUR cooperation.

The indirect influence of IPDC policy on E-IUR cooperation: enterprise innovation investment and regional IPLE. The IPDC policy can encourage enterprises to increase their investments in innovation (Jiang et al., 2023; Lv et al., 2023). It acts as a catalyst to allow enterprises to secure loans utilizing their IP assets as collateral. By leveraging property rights in patents, trademarks, and copyrights, enterprises can enhance their innovation capabilities by alleviating the financial barriers associated with pursuing innovative activities. The IPDC also plays a pivotal role in accelerating knowledge dissemination of knowledge through the establishment of innovation cooperation platforms and innovation incubation platforms. These initiatives foster collaboration among enterprises, universities, and research institutions, fostering an environment conducive to promoting effective cooperation and partnership in the field of innovation and research. The CNIPA-issued Index System for Evaluating Demonstration Cities clearly includes indicators such as “the amount and number of patent pledge financing” and “the amount and number of patent insurance coverage” and other indicators to measure the capacity to utilize urban IP. This has formed a strong guarantee for the R&D and innovation activities of enterprises with high capital investment and high technology content (Zhou and Guo, 2023; Jiang et al., 2023).

The high risk and uncertainty of the innovation activities of the enterprise leads to an increase in financial constraint (Wang et al., 2017). Developing economies lacking a mature institutional investor base and robust legal systems exhibit limited tolerance for innovation setbacks. The IPDC policy prioritizes offering financial services to enterprises and considers IP-based collateral financing a central focus (Song et al., 2023). With the gradual implementation of IPDC policy, IP pledge financing provides unprecedented opportunities for enterprises, allowing enterprises to use intellectual assets as collateral, easing financing constraints, and encouraging enterprises to invest in innovation. According to CNIPA, the total amount of funding of the patent right pledge in China has been expanding. The IPDC policy takes the financing of patent pledge to solve these constraints and promotes the investment of enterprise innovation by increasing the tolerance of banks to innovation failure.

In addition, increasing investments in innovation is conducive to consolidating the dominant position of enterprises in IUR cooperation and strengthening the deep integration of E-IUR cooperation. According to the resource dependence theory, the more the resource provider owns the key resources, the higher the dependence of the organization on the resource provider and the greater the power of the resource provider to the organization (Casciaro and Piskorski, 2005; Gulati and Sytch, 2007). The increased investment of enterprises in innovation can help overcome the symmetrical or reciprocal relationship among multiple innovation organizations in the IUR cooperation. Additionally, it can enhance the advantages of enterprises in accumulating professional knowledge in specific fields, building

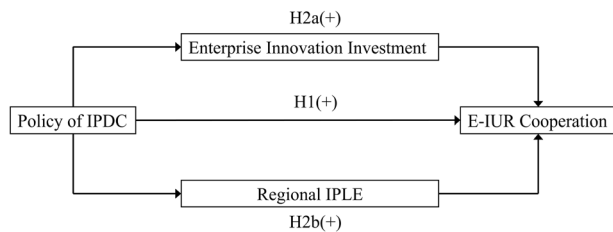


Fig. 1 Theoretical framework of the IPDC policy and E-IUR cooperation. The boxes represent variables, and arrows indicate the relationships between variables.

R&D platforms and increasing enterprise innovation investment. At the same time, the asymmetric dependence of innovative organizations such as universities and research institutions on enterprises will also be strengthened (Hoppmann, 2021), further consolidating the dominant position of enterprises in IUR cooperation. On the basis of the above analysis, this paper puts forward the following hypothesis.

Hypothesis 2a (H2a): *The IPDC policy has a positive effect on E-IUR cooperation by increasing enterprise innovation investment.*

The lack of effective enforcement mechanisms remains the predominant factor that contributes to inadequate IP protection in developing economies (Huang, 2017; Song et al., 2023). The IPDC policy is helpful in improving the regional IP law enforcement (regional IPLE). The IPDC policy can reduce the risk of infringement and increase the expected return of enterprise innovation investment by improving the efficiency of IP administrative enforcement, thus stimulating enterprises to engage in more innovative activities. IP protection ability is an important evaluation index of IPDC. The CNIPA issued Index System for Evaluating Demonstration Cities clearly includes such as “the number of cases of patent infringement disputes accepted”, “the number of closed cases of investigating and handling counterfeit patents”, and “the number of full-time law enforcement personnel to measure the IP protection capacity of the region”.

In addition, the promotion of regional IPLE will help consolidate the dominant position of enterprises in IUR cooperation and strengthen the deep integration of E-IUR cooperation. A set of IP laws has been enacted to enhance the judicial protection of IP rights, improving enterprises’ capacity to defend their rights. In doing so, the enforcement of IP serves as an effective mechanism to rectify market failures stemming from externalities like imitation and knowledge spillover. This legal framework contributes to a more robust protection of IP, mitigating challenges posed by unauthorized replication and the inadvertent diffusion of proprietary knowledge in the market (Albino-Pimentel et al., 2022). A city’s strong IP protection capabilities can ensure that the innovation property rights of enterprises are protected, thus maintaining the expected return on innovation (Lai et al., 2020). With guaranteed intellectual property rights, enterprises are more willing to disclose their innovative information, which attracts followers to purchase their IP rights. This enhances the R&D and innovation capabilities of enterprises, ultimately consolidating their dominant position in IUR cooperation and strengthening the deep integration of E-IUR cooperation. Based on the preceding analysis, this paper proposes the following hypothesis.

Hypothesis 2b (H2b): *The IPDC policy has a positive effect on E-IUR cooperation through promoting regional IPLE.*

The theoretical framework is shown in Fig. 1.

Research design

Sample selection and data sources. Our sample consists of Chinese-listed firms between 2008 and 2020. The period for sample selection from 2008 to 2020 is mainly due to two reasons. First, Chinese accounting standards changed after 2007. In 2007, firms implemented a new accounting system - “Accounting Standards for Business Enterprises (Order No. 33 of the Ministry of Finance).” The adoption of new accounting standards can have a certain impact on financial statistical accounting. The methods of financial data statistics may vary, so the period is chosen to start from 2008 to account for these changes. Second, although the most recent data for the SSTP Awards is updated to 2023, the awards are suspended in 2021 and 2022, so our data only covers 2020.

Our data come mainly from the WIND database and the China Stock Market and Accounting Research (CSMAR) database, the National Office for Science and Technology Awards (NOSTA) official website (<https://www.nosta.gov.cn/p/en/awards/gjksjsbj/index.html>), and CNIPA. The WIND database and the CSMAR database (Zhou et al. 2017) provide financial and attribute data such as firm age, firm size, the state ownership of the firm, ownership concentration, cash flow, firm leverage, investment to total asset ratio, and return on assets variables (Jia et al., 2019). The data on E-IUR cooperation come from the NOSTA. The official website of NOSTA annually showcases the prestigious list of SSTP awards, covering ~200 items. The data of IPDC policy come from CNIPA, excluding county-level cities and municipalities directly under the Central Government. Over the past decade, considerable effects have been undertaken by the Chinese government to advance the construction of IPDCs, resulting in the establishment of 77 IP demonstration cities to date. We dropped jurisdiction rather than the city itself as it belongs to the scope of the demonstration of firms and the county-level demonstration city. The cities where the sample enterprises are located involve 60 demonstration cities, see Fig. 2.

The specific data processing procedures are as follows: First, we compiled the winners of the SSTP Awards on the official website, including all awarded organizations. To standardize the sample data, we addressed enterprise name changes by using the Unified Social Credit Code for identification. We match the data obtained from the WIND, CSMAR database, and NOSTA by the stock names and names of listed companies. Additionally, we matched the registered city of each enterprise to ascertain whether it is designated as an Intellectual Property Demonstration City (IPDC) from CNIPA. A total number of 26,496 firm-year observations from 3,152 firms is obtained after excluding ST, *ST, and PT firms; firms in the financial industries, firms with a debt-to-asset ratio greater than 1, and jurisdiction rather than the city itself belong to the scope of the demonstration of firms and include the county-level demonstration city of firms, and firms that delisted during the 2008 to 2020 timeframe. All continuous variables have been winsorized at the 1st and 99th percentiles to mitigate the influence of outliers.

Empirical strategy. We regard the IPDC policy as a quasi-natural experiment and construct a staggered DID model to investigate the influence of the IPDC policy on the deep integration of E-IUR cooperation. Specifically, we estimate the following fixed-effects ordinary least squares regressions:

$$EIURC_{it} = \beta_0 + \beta_1 Treat_IPDC_{it} \times post_{it} + \beta_2 \bar{X}_{it} + ind_{it} + \sigma_i + \mu_t + \varepsilon_{it} \quad (1)$$

where i indexes the enterprise and t indexes time. $EIURC_{it}$ is the dependent variable (the deep integration of E-IUR cooperation),

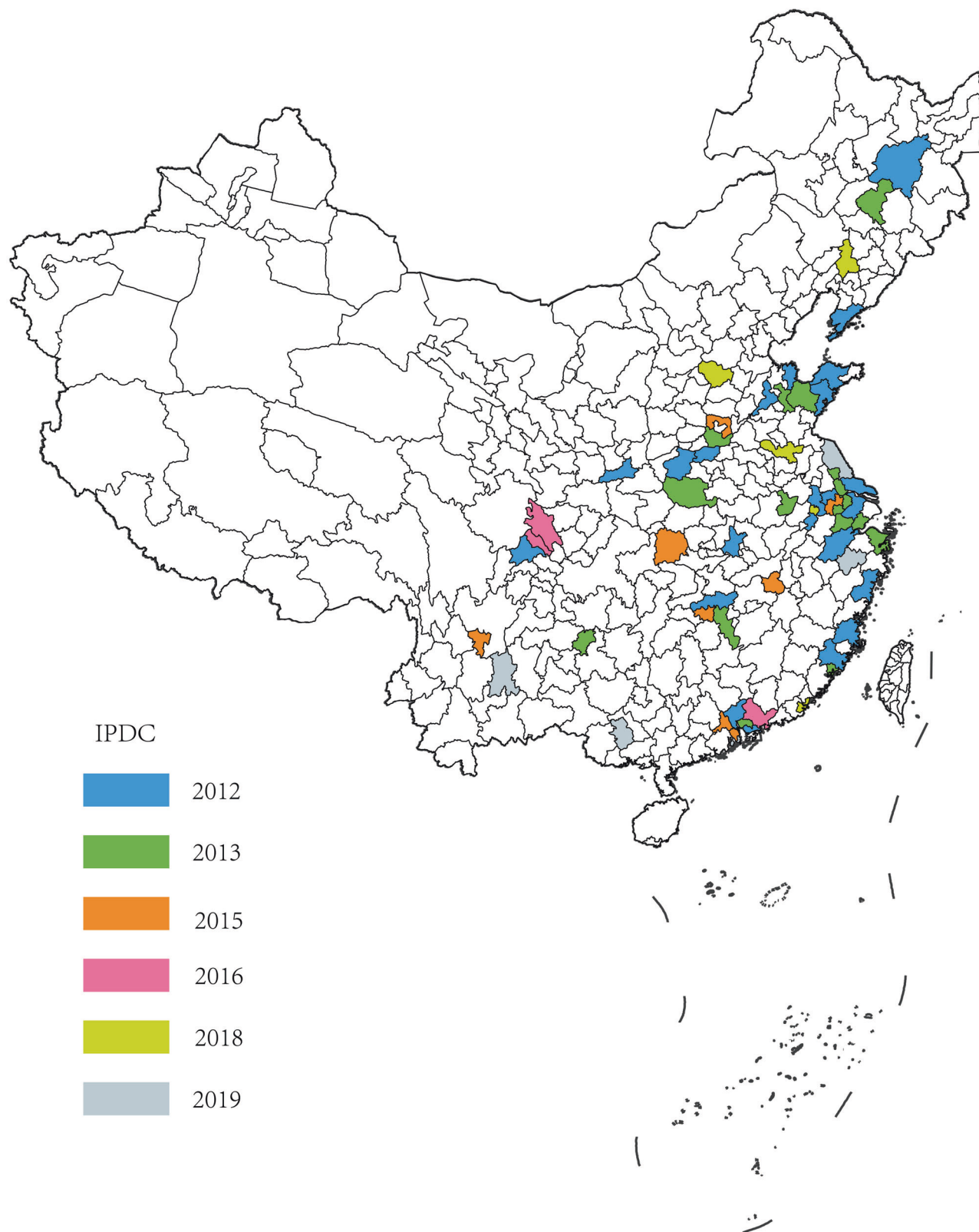


Fig. 2 Summary of IPDC involved in sample enterprises. The geographical distribution of the 60 demonstration cities where the sample enterprises in our study are located.

σ_i is the firm fixed effects, μ_t is the year fixed effects, and ε_{it} is an error term. $\bar{X}_{it}$ is a vector of all control variables. Because the specification includes enterprise fixed-effects and year fixed-effects, it is not necessary to include treatment and time variables.

The effect of the IPDC policy on the deep integration of E-IUR cooperation is captured by β . A positive and significant β suggests that the IPDC policy exerts a positive effect on the deep integration of E-IUR cooperation.

Variables

Dependent variable. Our dependent variable is the deep integration of E-IUR cooperation (EIURC). Innovation can be conceptualized as a two-stage process (Zhang et al., 2014). The first stage involves basic research and invention, while the second stage centers on commercialization or adaptation of innovations for commercial use. It's important to note that metrics such as patent counts or patent citations primarily capture outcomes from the invention phase of innovation, indirectly reflecting aspects of the commercialization process. We define the deep integration of E-IUR cooperation effectiveness based on its capacity to generate impactful commercial outputs. To gauge this, we employ the attainment of innovation awards, such as the SSTP Award, as a proxy (Blau and McKinley, 1979; Zhang et al., 2014).

The SSTP Award is given to organizations that have significantly contributed to applying and disseminating advanced research findings or completing major R&D projects, plans, and programs. The SSTP Award recognizes outstanding technological innovation with advanced technical and economic indicators. It has shown significant economic, social, and ecological benefits after application and promotion, alongside notable contributions to national security. Moreover, it plays a pivotal role in advancing scientific and technological progress within the industry. The SSTP Award can reflect the position of enterprises within the SIS (Malerba, 2002) and their engagement in collaborative innovation with key stakeholders from industries, universities, and research institutions. We select the SSTP Award to measure the deep integration of E-IUR cooperation, mainly based on the following considerations. First, the data is authoritative. As one of the highest awards for science and technology in China, the SSTP Award is highly prestigious and represents high-level and high-quality technological innovation achievements. The SSTP Award honors individuals or organizations that have made remarkable contributions to promoting the advancement of S&T and economic and social development through the implementation and application of innovative S&T achievements. Second, the data is accessible. The SSTP Award is published every year, with data being open and transparent, and the order of the main completion organizations can be observed, which can represent the degree of the deep integration of E-IUR cooperation. Specifically, the enterprise won the SSTP Award with the first completion organization as E-IUR cooperation with a high degree of deep integration, which is recorded as 2. The enterprise won the SSTP Award with the non-first completion organization, and the degree of deep integration in E-IUR cooperation is low, which is recorded as 1. If the enterprise has not won the SSTP Award, it is the degree of deep integration of E-IUR cooperation that will be considered as 0.

Independent variable. We use DID regressions to test our hypotheses (Guo et al., 2016; Li et al., 2022). In the context of our regression analyses, our focus lies on assessing the significance level and magnitude of the difference estimator within the IPDC, specifically the interaction term representing the combination of treatment and post-period (DID_IPDC). To put it differently, our independent variable in this study is the interaction effect of treatment and post-period, emphasizing the joint influence rather than considering each factor in isolation. Treatment receives a value of "1" for enterprises located in demonstration cities (urban areas) (treatment groups) and "0" for the rest (control groups). The post period is coded "1" when the city where the enterprise is registered is recognized as a demonstration city of IP in a certain year and "0" for all previous years.

Control variables. Several control variables of firm age, firm size, the state ownership of the firm, ownership concentration, cash

flow, firm leverage, investment to total asset ratio, and return on assets variables are included based on the literature. Firm age (FA) is considered because younger firms may prioritize innovation over older ones (Luo et al., 2017). Firm age is defined as the number of years since the firm was founded. Because larger firms can access more innovation resources to invest, we control for firm size (FS), measured by the logarithm of the total assets (Choi and Lee, 2018). We determine the state ownership of firms (SO) based on the proportion of state-owned shares. Firms with state-owned shareholding exceeding 50% are classified as SOEs, while those below are classified as non-SOEs (Kafourous et al., 2015; Zhou et al., 2017). The operation of firms' ownership is denoted as 1 for SOE and 0 for non-SOE. We also included ownership concentration (OC) represented by the proportion of shares held by the firm's largest shareholder (Gedajlovic and Shapiro, 1998; Li et al., 2010). Cash flow (CF) is assessed using the ratio of net operating cash flow to total assets. We control for firm leverage (Lev) in terms of the debt-to-assets ratio (He et al., 2022). The Investment to total asset ratio (Invt), is measured by the proportion of total assets allocated to the acquisition and construction of fixed assets, intangible assets, and other long-term assets at year-end. We incorporate financial performance, specifically the return on assets (ROA), as successful firms with greater financial capacity are more capable of allocating resources towards innovation endeavors.

Results and analysis

Descriptive statistics. Tables 1 and 2 provide an overview of the descriptive statistics and the correlations among the variables considered in our analyzes. Table 1 displays the descriptive statistics of all variables. The average degree of E-IUR cooperation (EIURC) is 0.0130, and the standard deviation is 0.1260, which shows that there are great differences among enterprises. The average value of the policy for IPDC (DID_IPDC) is 0.3970, which indicates that 39.70% of the samples were covered by the policy for IPDC during the sample period. Table 2 displays the correlations of the dependent variables, independent variable, and control variables. We perform a common variance inflation factor (VIF) test, confirming that there is no multicollinearity in the data. The maximum VIF is 1.59 and the average VIF is 1.25, indicating that all VIF values are lower than the commonly used threshold of 10.

Data from the NOSTA's official website indicates that ~200 to 300 enterprises secure the SSTP Award each year. As evident from Table 3, in 2014, the organization led by the enterprises won the SSTP Award accounting for the highest proportion of all winning programs, reaching 36.42%. However, by 2020, it only represented 21.49% of all award-winning programs. In summary, the proportion of awards won by enterprises as the primary entity

Table 1 Variables descriptive statistics.

No.	Variables	Obs	Mean	SD	Min	Max
1	EIURC	26496	0.0130	0.1260	0.0000	2.0000
2	DID_IPDC	26496	0.3970	0.4890	0.0000	1.0000
3	FA	26496	2.7880	0.3460	1.3860	3.9890
4	FS	26496	3.6810	1.2810	0.0048	10.2200
5	SO	26453	0.3870	0.4870	0.0000	1.0000
6	OC	24301	35.2800	15.1400	0.0000	100.0000
7	CF	26496	0.0496	0.0716	-0.1690	0.2520
8	Lev	26496	0.4290	0.2080	0.0506	0.8860
9	Invt	26484	0.0510	0.0480	0.0002	0.2300
10	ROA	26496	0.0412	0.0586	-0.2210	0.2050

Source(s): Authors' own work.

Table 2 Correlation analysis of variables.										
Variables	1	2	3	4	5	6	7	8	9	10
EIURC	1.0000									
DID_IPDC	−0.0020	1.0000								
FA	−0.0260***	0.3180***	1.0000							
FS	0.1100***	0.0360***	0.1110***	1.0000						
SO	0.0200***	−0.1620***	−0.0920***	0.3200***	1.0000					
OC	0.0230***	−0.0430***	−0.1110***	0.1780***	0.2190***	1.0000				
CF	0.0120*	0.0210***	0.0040	0.0480***	−0.0050	0.0930***	1.0000			
Lev	0.0360***	−0.0640***	−0.0100	0.4690***	0.3060***	0.0510***	−0.1570***	1.0000		
Invt	0.0130**	−0.0600***	−0.1380***	−0.0360***	−0.0660***	0.0410***	0.1840***	−0.0620***	1.0000	
ROA	0.0020	0.0210***	−0.0460***	−0.0390***	−0.1330***	0.1230***	0.3820***	−0.3650***	0.1490***	1.0000

Standard errors in parentheses **p* < 0.1, ***p* < 0.05, ****p* < 0.01; Source(s): Authors' own work.

Table 3 E-IUR Cooperation—Taking the SSTP award as an example.				
Year	Number of enterprises	Number of Led-enterprises	Total of awards	Proportion of E-IUR
2008	255	53	179	29.12%
2009	283	60	137	27.03%
2010	320	62	156	28.97%
2011	331	62	154	28.44%
2012	261	49	185	30.82%
2013	227	46	175	34.33%
2014	297	55	184	36.42%
2015	286	45	192	32.61%
2016	267	29	182	22.48%
2017	287	35	222	27.13%
2018	276	31	214	23.13%
2019	352	27	218	18.62%
2020	280	26	159	21.49%

Source(s): Authors' own work.

is relatively low and exhibits a downward trend, indicating that enterprises are not in a dominant position in technological innovation. This suggests that in the NIS of China, the level of E-IUR cooperation is low.

Results for hypothesis testing. The hypotheses are tested with the fixed effects models. We present the findings of estimating Eq. (1) in Table 4. It reports the estimation results of DID regression estimation for the IPDC policy on the degree of E-IUR cooperation. In Model 1, only the IPDC policy (DID_IPDC) is added, and its estimation coefficient is 0.0052, which is significant at the level of 10%, indicating that the IPDC policy is helpful in strengthening the deep integration of E-IUR cooperation. This estimate constitutes the central finding of the paper. Assuming that the regression can be interpreted causally, it suggests that the implementation of IPDC policy enhances the deep integration of E-IUR cooperation in the region compared to the control group. Adding all the control variables in Model 2, the coefficient of the IPDC policy (DID_IPDC) is still positive, and it is significant at the level of 5%, which shows that the IPDC policy has significantly strengthened the deep integration of E-IUR cooperation under the premise of other conditions unchanged and verified the hypothesis 1. From the economic point of view, taking the results of Model 2 as an example, compared with those enterprises in non-IPDC, when a city where an enterprise is located is recognized as an IP demonstration city, the degree of E-IUR cooperation has increased by 0.0094. After controlling for other

Table 4 Benchmark regression results.		
Variables	Model 1	Model 2
DID_IPDC	0.0052* (0.0031)	0.0094** (0.0044)
FA		−0.0340** (0.0135)
FS		0.0037* (0.0021)
SO		−0.0048 (0.0061)
OC		0.0000078 (0.0001)
CF		0.0040 (0.0144)
Lev		0.0026 (0.0086)
Invt		−0.0054 (0.0230)
ROA		0.0047 (0.0198)
Industry	No	Yes
_cons	0.0110*** (0.0014)	0.1050 (0.0708)
N	26198	24108
adj. R ²	0.1220	0.1210

Standard errors in parentheses **p* < 0.1, ***p* < 0.05, ****p* < 0.01; Source(s): Authors' own work.

variables affecting IPDC policy on the degree of E-IUR cooperation, the previous conclusions (hypothesis 1) remain true.

Robustness test
Parallelism trend test. The premise of using staggered DID is to meet the hypothesis of parallel trend, which requires that there be no systematic difference in the changing trend of the explained variables between the control group and the treatment group before the enterprise enjoys the IPDC policy, that is, there is a parallel trend. In this paper, the event analysis method is used to test the parallelism trend, and the specific model is as follows:

$$EIURC_{it} = \beta_0 + \sum_{k=-5}^{k=5} \beta_k D_{it}^k + \beta_2 \bar{X}_{it} \mu_t + ind_{it} + \sigma_i + \varepsilon_{it} \quad (2)$$

Wherein, the meaning of $EIURC_{it}$, ind_{it} , μ_t , σ_i , ε_{it} is consistent with formula (1). D_{it}^k represents the dummy variable of the time when the enterprise's area becomes IP demonstration city. If $t - year_i \leq -5$, then $D_{it}^{-5} = 1$, otherwise 0; if $-5 \leq t - year_i \leq 5$, then $D_{it}^k = 1$, otherwise 0; if $t - year_i \geq 5$, then $D_{it}^5 = 1$, otherwise 0. To avoid the impact of multicollinearity, a year before the

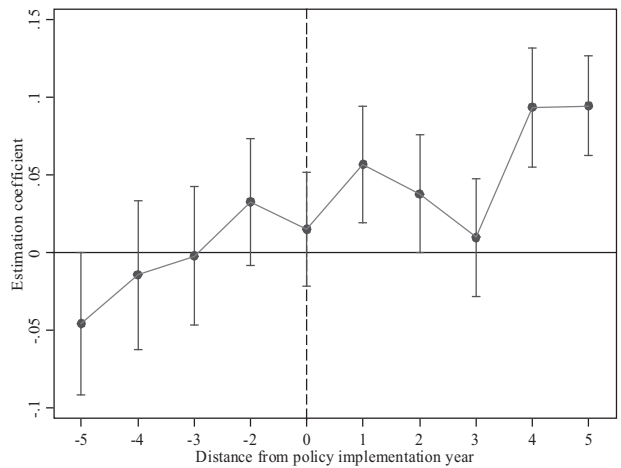


Fig. 3 Diagram for the parallel trend and dynamic effect. The estimated coefficient and its 95% confidence interval for the policy variable related to IPDCs show no significant difference in E-IUR cooperation before the policy implementation, thus supporting the parallel trend hypothesis.

implementation of the policy is used as a benchmark for regression.

Figure 3 shows the estimated value and the 95% confidence interval of the policy variable coefficient of the IPDC in Eq. (2). The horizontal axis represents the year before and after the implementation of the policy, and the vertical axis represents the difference effect coefficient of the degree of E-IUR cooperation. The point in Fig. 2 is the value of the estimation coefficient and the real line segment combing through the point is the 95% confidence interval of the estimation coefficient. In specifications where we use E-IUR cooperation as the dependent variable, the coefficients for the before-policy periods (from 5 to 2 years before) consistently show insignificance. Therefore, before the region where the enterprise is located becomes the IPDC, the differences in the explained variables cannot reject the original hypothesis; that is, there is no systematic difference in the degree of E-IUR cooperation, which satisfies the parallel trend hypothesis.

PSM-DID. Although the DID regression method can effectively alleviate endogenous problems, the potential sample selection bias may still interfere with the accuracy of the estimation results. To alleviate the problem of sample selection bias, this paper uses the propensity matching score method (PSM) to match the samples similar to the treatment group from the original control group as the new control group, and then uses the DID method to test the two groups of samples, to eliminate the interference caused by the differences between the two groups of sample policies before implementation, and can more accurately estimate the impact of the IPDC policy on the degree of E-IUR cooperation. In this paper, the tendency score of each sample is calculated by the Probit model. We include all the variables used in the benchmark model as control variables. Then, the treatment group and the control group are matched in two ways: K nearest-neighbor one-to-four matching and radius matching in calipers. In Models 1 and 2, we used K nearest neighbor one-to-four matching and DID, the estimation coefficients ($\beta_1 = 0.0052, p < 0.1$; $\beta_2 = 0.0060, p < 0.1$) show that the implementation of IPDC policy could increase the deep integration of E-IUR cooperation in the region compared to the control group. In Models 3 and 4, employing radius matching in calipers and DID, the estimation coefficients ($\beta_1 = 0.0069, p < 0.1$; $\beta_2 = 0.0068, p < 0.1$) indicate that IPDC policy still

Table 5 Robustness test (PSM-DID).

Variables	K nearest neighbor matching in caliper ($k = 4$)		Radius matching(0.05)	
	Model 1	Model 2	Model 3	Model 4
DID_IPDC	0.0059 [*] (0.0034)	0.0060 [*] (0.0034)	0.0069 ^{**} (0.0033)	0.0068 ^{**} (0.0033)
FA		-0.0312 ^{**} (0.0141)		-0.0340 ^{**} (0.0135)
FS		0.0033 (0.0022)		0.0037 [*] (0.0021)
SO		-0.0067 (0.0064)		-0.0048 (0.0061)
OC		-2.4200E-06 (0.0001)		7.8800E-06 (0.0001)
CF		0.0037 (0.0151)		0.0040 (0.0144)
Lev		0.0032 (0.0090)		0.0026 (0.0086)
Invt		0.0059 (0.0243)		-0.0054 (0.0230)
ROA		-0.0041 (0.0207)		0.0047 (0.0198)
Industry	No	Yes	No	Yes
_cons	0.0114 ^{***} (0.0015)	0.1050 (0.0760)	0.0110 ^{***} (0.0015)	0.1050 (0.0708)
N	21689	21689	24107	24107
adj. R ²	0.1300	0.1300	0.1210	0.1210

Standard errors in parentheses ^{*} $p < 0.1$, ^{**} $p < 0.05$, ^{***} $p < 0.01$; Source(s): Authors' own work.

enhances the deep integration of E-IUR cooperation in the region relative to the control group. In this paper, the matched samples are used for regression, and the results show that the IPDC policy is conducive to enhancing the impact on the deep integration of E-IUR, which shows that the previous conclusions of this paper are still valid and the specific results are shown in Table 5.

Treatment of endogenous problems. The choice of IPDC is not random, which may lead to selection bias. The DID model can effectively control non-observational factors that do not change with time, but it can still not completely deal with endogenous factors. PSM-DID can avoid some problems to some extent, but it still can't be completely solved. The two-stage treatment effects model can help to resolve endogeneity resulting from sample selection (Maddala, 1983). In the first stage, Probit model is used to estimate the selection equation of whether the city where the enterprise is located becomes an IPDC. The ability of the city to attract foreign investment (Foreign_invest) is used as an instrument variable to enter the regression model, and the number of foreign-invested enterprises in the city in that year is measured logarithmically. In the period of economic transformation, local officials compete in economic growth, resource competition and central government support to get promotion opportunities. The development of local economy and the application for the title of local IPDC are one of the performance evaluations of local governments. Therefore, more ambitious local governments are more likely to strive for the selection of IPDC. The degree of local government efforts to attract foreign investment in the process of economic development is an indicator to measure the ambition of local government (Guo et al., 2016). Therefore, we expect that this instrument variable is positively related to the probability that the city will become an IPDC. Then, the inverse Mills ratio (IMR) calculated in the first stage is substituted into the regression

model in the second stage to control the potential endogenous problems (Certo et al., 2016).

Table 6 lists the regression results of the two-stage treatment effects model. Model 1 shows that there is a significant positive correlation between a city’s ability to attract foreign investment and whether it becomes an IPDC ($\beta = 0.0527$, $\rho < 0.01$). Model 2 shows that after considering the possible sample selection bias, the positive correlation between the IPDC policy and degree of E-IUR cooperation still holds ($\beta = 0.0075$, $\rho < 0.05$). In summary, after considering possible sample selection bias, the results are consistent with the baseline model, and the research conclusion is robust.

Mechanism analysis. Our main set of analyzes investigated how the IPDC policy influences the degree of E-IUR cooperation. According to the logic of theoretical analysis, this paper tests the two paths of increasing the enterprise innovation investment and enhancing the regional IPLE respectively. Enterprise innovation investment (RD) is measured by the ratio of R&D expenditure to operating income in the current year; the regional IPLE (RIPLE) is measured by the number of cases of IP in the city where the enterprise is located in that year and is logarithmized. This paper refers to the existing research three-step mediation effect model to verify the two channels of enterprise innovation investment and regional IPLE (Baron and Kenny, 1986).

$$EIURC_{it} = \alpha_0 + \gamma Treat_IPDC_{it} \times post_{it} + \beta_2 \bar{X}_{it} + \mu_t + ind_{it} + \sigma_i + \varepsilon_{it} \tag{3}$$

Table 6 Endogeneity test results by using the two-stage treatment effects model.		
	1st Stage IPDC	2st Stage EIURC
Foreign_invest	0.0527*** (0.0047)	
FA	0.3360*** (0.0271)	−0.0404*** (0.0152)
FS	−0.0021 (0.0079)	0.0038* (0.0021)
SO	−0.2900*** (0.0187)	0.0016 (0.0092)
OC	0.0022*** (0.0006)	−0.00004 (0.0001)
CF	−0.2120* (0.1264)	0.0087 (0.0152)
Lev	0.1140** (0.0500)	0.0004 (0.0089)
Invt	0.4060** (0.1776)	−0.0136 (0.0246)
ROA	0.4810*** (0.1706)	−0.0055 (0.0227)
Industry	−0.0833*** (0.0036)	−0.0010 (0.0126)
DID_IPP		0.0075** (0.0033)
IMR		−0.0374 (0.0401)
_cons	−0.3320*** (0.0858)	0.1380* (0.0796)
N	24286	24108
Pseudo R ² /adj. R ²	0.0347	0.1210

Standard errors in parentheses *p < 0.1, **p < 0.05, ***p < 0.01; Source(s): Authors’ own work.

$$M_{it} = \alpha_1 + \theta Treat_IPDC_{it} \times post_{it} + \beta_2 \bar{X}_{it} + \mu_t + ind_{it} + \sigma_i + \varepsilon_{it} \tag{4}$$

$$EIURC_{it} = \alpha_0 + \delta Treat_IPDC_{it} \times post_{it} + \varphi M_{it} + \beta_2 \bar{X}_{it} + \mu_t + ind_{it} + \sigma_i + \varepsilon_{it} \tag{5}$$

Wherein, the meaning of $EIURC_{it}$, ind_{it} , μ_t , σ_i , ε_{it} is consistent with Eq. (1). M_{it} represents the mediating variables. The total effect of the IPDC policy is γ . The direct effect of the IPDC policy is δ . The effect of the mediating variables is φ . If γ is statistically significant, and δ , θ and φ is statistically significant too, it means that the mediating variable M_{it} is a complete intermediary. Otherwise, the mediating effect does not exist.

In Models 1 and 2 of Table 7, we explore the mediating effect of the enterprise innovation investment about the IPDC policy on the degree of E-IUR cooperation. Model 1 shows that the coefficient of the IPDC policy (DID_IPDC) is 0.1360, which is significant at the level of 5%, indicating that the IPDC policy has significantly improved the enterprise innovation investment. In model 2, the coefficient of enterprise innovation investment is 0.1510, which is significant at the level of 1%, while the coefficient of the IPDC policy (DID_IPDC), is 0.0105, which is not significant, indicating that the enterprise innovation investment is a complete mediating effect, indicating that the IPDC policy has strengthened the degree of E-IUR cooperation by increasing the enterprise innovation investment, and the hypothesis 2a has been verified.

In Models 3 and 4 of Table 7, we explore the mediating effect of the regional IPLE where the enterprise is located about the IPDC policy on the degree of E-IUR cooperation. Model 3 shows that the DID_IPDC coefficient of the IPDC policy is 0.5340, which is significant at the level of 1%, indicating that the IPDC policy has significantly improved the regional IPLE (RIPLE) where the enterprise is located. In model 4, the coefficient of the regional IPLE (RIPLE) is 0.0924, which is significant at the level of 5%, while the coefficient of the IPDC policy (DID_IPDC) is 0.1460, which is not significant, indicating that the regional IPLE is a complete mediating effect, indicating that the IPDC policy has strengthened the degree of E-IUR cooperation by enhancing the regional IPLE, and the hypothesis H2b has been verified.

Further analysis. The CNIPA introduced the first group of 127 state-level IP demonstration enterprises (IPDE). Subsequently, from 2015 to 2019, for five uninterrupted years, the list of IPDE was annually updated, encompassing a cumulative total of 993 enterprises, among which 278 are publicly listed. Utilizing the same dataset, comprising the same years and control variables, we delved into the influence of IPDE policy on the degree of E-IUR cooperation, as evident in Table 8.

In Model 1, we focused solely on the policy of IPDE, yielding an estimated coefficient of 0.0171, significant at the 1% level. This suggests that the policy of IPDE contributes substantially to the enhancement of the degree of E-IUR cooperation. When we incorporated all control variables in Model 2, the estimated coefficients for IPDE remained positive and significant at the 1% level, indicating that the policy of IPDE promotional effect on the degree of E-IUR cooperation persists even under varying conditions. Our empirical findings reveal that the policy of IPDE is more effective than IPDC policy in advancing the degree of E-IUR cooperation. Future studies can investigate in more depth the impact of both IPDE and IPDC policies on the degree of E-IUR cooperation.

Table 7 Mediation effect test.

Variables	Model 1 RD	Model 2 EIURC	Model 3 RIPLE	Model 4 EIURC
DID_IPDC	0.1360** (0.0589)	0.0105 (0.1426)	0.5340*** (0.0240)	0.1460 (0.1494)
RD		0.1510*** (0.0235)		
RIPLE				0.0924*** (0.0280)
FA	−0.146 (0.2954)	−1.193*** (0.2144)	−0.770*** (0.0987)	−1.106** (0.4536)
FS	−0.0243 (0.0405)		−0.0462*** (0.0150)	
SO	0.0248 (0.1197)	−0.171 (0.1590)	−0.127*** (0.0448)	0.00602 (0.1420)
OC	−0.00251 (0.0026)	−0.000683 (0.0043)	0.00132 (0.0010)	0.00505 (0.0041)
CF	−0.920*** (0.2694)	1.638 (1.1513)	0.00760 (0.1055)	1.760* (0.9798)
Lev	−2.000*** (0.1598)	0.599 (0.4614)	−0.0406 (0.0628)	2.067*** (0.3600)
Invt	3.782*** (0.4126)	−0.800 (1.5158)	−0.176 (0.1686)	1.511 (1.3101)
ROA	−6.099*** (0.3489)	0.329 (1.6048)	−0.184 (0.1452)	2.182 (1.3830)
Industry	Yes	Yes	Yes	Yes
_cons	6.4220*** (1.2146)	0.5490 (0.7097)	7.8850*** (0.5152)	−1.0440 (1.2405)
N	18361	18580	23796	23972
adj. R ²	0.8040		0.8910	
LR chi2		76.8500		73.1400

Standard errors in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; Source(s): Authors' own work.**Table 8 Intellectual property demonstration enterprise policy effect.**

Variables	Model 1	Model 2
DID_IPDE	0.0171*** (0.0058)	0.0161*** (0.0059)
FA		−0.0403*** (0.0124)
FS		0.00235 (0.0018)
SO		−0.000731 (0.0056)
OC		−0.0000927 (0.0001)
CF		0.00469 (0.0132)
Lev		0.00473 (0.0078)
Invt		−0.00317 (0.0213)
ROA		0.00924 (0.0177)
Industry		−0.00789 (0.0181)
_cons	0.0135*** (0.0007)	0.135** (0.0534)
N	29767	29487
R ²	0.228	0.230

Standard errors in parentheses ** $p < 0.05$, *** $p < 0.01$; Source(s): Authors' own work.

Conclusion and discussion

The impact of IP on enterprise innovation or IUR cooperation is a widely debated issue, especially in periods of institutional transition such as China. Based on the quasi-experiment of the IPDC policy, this paper takes the listed firms in China from 2008 to 2020 as the research sample, and uses the DID regression model to analyze the influence of the IPDC policy on the deep integration of E-IUR cooperation, and further explores the influence mechanism of the IPDC policy on the deep integration of E-IUR cooperation from two paths: enterprise innovation investment and regional IPLE. The results show that the IPDC policy can significantly strengthen the deep integration of E-IUR cooperation. Mechanism analysis shows that the IPDC policy can strengthen the deep integration of E-IUR cooperation through two intermediary paths: increasing the enterprises' innovation investment and enhancing the regional IPLE. Our findings align with previous findings that IP could promote IUR collaboration (Okamuro and Nishimura, 2013; Shi et al., 2020; George and Tarr, 2023), further deepening the depth of IUR collaboration. We examine not only the asymmetric dependence in IUR cooperation but also the innovation process measuring by SSTP Award, which is more oriented towards commercialization than mere patenting.

Our above findings contribute to the IP and IUR cooperation literature in several ways. First, we use the SSTP Award as an indicator of the effectiveness of IUR cooperation. Previous studies focus mainly on the discussion of whether IP policy has a good effect on R&D investment or patent application (Brüggemann et al., 2016; Fang et al., 2017; Song et al., 2023), ignoring the

innovation commercialization process. Our emphasis is on recognized innovations that have been certified by third-party evaluations, highlighting technological advancements, substantial economic advantages, and advances in the industry, thus establishing a standard for expectations in E-IUR collaboration.

Second, from the perspective of asymmetric dependence, this paper investigates the dominant-subordinate relationship among multiple innovation organizations in IUR cooperation, measures the dominant degree of E-IUR cooperation compared to universities, research institutions, and other innovation organizations, and deepens the cooperation research in IUR. While the existing literature has predominantly concentrated on the impact of IP protection on innovation (Brüggemann et al., 2016; Song et al., 2023; Zhou and Guo, 2023; Zheng et al., 2023; Song et al., 2024), it has often overlooked the complex dynamics driving collective innovation outcomes. This study challenges the assumption of an egalitarian relationship among various innovation organizations within the IUR framework, addressing overlooked power imbalances arising from resource dependence dynamics (Song, 2013; Hoppmann, 2021).

Finally, we put IP protection and the IUR innovation cooperation into the same framework and explores the influence of IP protection on the relationship among multiple innovation organizations in the SIS, enriching research on the innovation effect of IP protection (Bondarev, 2018; Song et al., 2023; Nguyen et al., 2023). Existing research focuses on the impact of IP rights on R&D investment and patent applications (Brüggemann et al., 2016; Fang et al., 2017; Song et al., 2023). Specifically, we emphasize the significant value of IP protection for fostering innovations with substantial commercialization potential. Furthermore, our findings offer strategic insights into enhancing deep integration within the E-IUR cooperation model, especially from the perspective of the institutional environment. This research not only provides empirical evidence of the benefits derived from IPDC policy but also offers actionable guidance for policymakers and business leaders.

The conclusion of this paper has important implications for the construction of IPDC and for strengthening the deep integration of E-IUR cooperation. First, the government should refine and summarize the experience in the construction of IPDC, form several practices featuring IP rights, create some demonstration benchmarks for IP protection, and then promote replication throughout the country to promote the overall level of IP protection. And the government should issue relevant policies to encourage and support more cities to participate in the construction of IPDC. The government should conduct continuous monitoring and evaluation of the IP protection work in demonstration cities, identify problems on time, and take corresponding measures to improve it, to ensure its leading and driving role. Second, the government should standardize and regulate the behavior of IP ownership, information data confidentiality, and benefit distribution in IUR cooperation, and promote the deep integration of E-IUR. In addition, the government should establish a complete set of norms for the cooperation of IUR, and clarify the responsibilities, rights, and obligations of all parties in the cooperation process, as well as the cooperation process, communication mechanism, etc., to ensure a smooth progress of the cooperation. In the process of cooperation of IUR, if there is any dispute, all parties should establish an effective dispute resolution mechanism, including negotiation, mediation, arbitration, etc., to ensure that the dispute is resolved in a timely and proper manner.

The findings of this study offer critical insights for enterprises looking to enhance their role in the NIS through the E-IUR cooperation model. First, it underscores the importance of

robust IP rights protection, suggesting that enterprises must actively engage with IPDC policy to safeguard their innovations. Enterprises should invest in understanding and leveraging IP laws to foster a culture of innovation and to protect their competitive edge. Second, the study highlights the need for continuous enterprises innovation investment, indicating that increased R&D spending can deepen E-IUR cooperation. Managers should prioritize budget allocations for R&D to drive technological advancements and to strengthen collaborative efforts with universities and research institutions. Third, the downward trend in E-IUR cooperation suggests that enterprises must be proactive in seeking partnerships and creating collaborative platforms that can sustain and enhance innovation. This may involve developing strategic alliances, joint ventures, or shared research initiatives. Lastly, the empirical evidence of study on the influence of IPDC policy on the integration of E-IUR cooperation provides a clear direction for policymakers and corporate leaders to optimize the basic system of IP. Enterprises should advocate and participate in the refinement of IP institutions to improve the overall efficiency of the NIS, thus contributing to the growth and competitiveness of emerging economies.

Data availability

The data in Dataset 1 are publicly available from WIND database, CSMAR database, and NOSTA (<https://www.nosta.gov.cn/pc/en/index.shtml>). The study used no special computer code.

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Competing interests

The authors declare no competing interests.

Additional information

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